

Artificial Intelligence and Machine Learning (AIML) – Project

Names:

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Problem Statement: VibeSync: AI Music Mood Recommendation and Smart Music Player

Title:

" VibeSync: AI Music Mood Recommendation and Smart Music Player "

Abstract:

VibeSync project aims to develop a personalized music recommendation system that dynamically suggests songs based on the user's mood and preferences. Utilizing machine learning algorithms, the system analyzes audio features, emotional tone, and user behavior to recommend music that matches the user's current mood and listening habits.

Dataset:

1. Title: Music Library Dataset

Features and Workflow:

1. Data Collection & Feature Extraction:

- Gather data from music libraries, including audio features such as tempo, rhythm, and key, as well as metadata (genre, artist, etc.).
- Use emotion detection techniques (speech recognition, facial expression analysis) or user inputs (manual selection) to assess mood.

2. Model Training:

- **Employ algorithms like Random Forests, Support Vector Machines (SVM), or Deep Neural Networks (DNNs) to train mood-based music recommendation models.**
- **Integrate collaborative filtering techniques for improved user preference accuracy.**

3. Recommendation Engine:

- **The engine will use extracted audio and behavioral data to recommend tracks that align with the user's mood (happy, calm, energetic, etc.).**
- **Real-time updates ensure that song suggestions adjust dynamically as user preferences evolve or the mood changes.**

4. AI-Integrated Music Player:

- **Seamlessly incorporate the recommendation engine into an AI-powered music player that auto-generates playlists and provides voice-assisted commands.**
- **Additional AI features include smart volume adjustment, transitions between songs, and personalized notifications based on user patterns.**

Expected Outcome:

The system will enhance the listening experience by offering mood-based recommendations, fostering music discovery, and building playlists tailored to individual user preferences. The AI music player, enriched with intelligent features, will transform how users interact with their music collections. Thus, this project aims to provide users with the most seamless musical experiences.